

Title: APT Profiling and Detection in Critical Infrastructure using Deception Tech

Group No : 10

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Problem Statement :

Advanced Persistent Threats (APTs) bypass traditional, signature-based defenses in critical infrastructure, using stealthy tactics that mimic legitimate user activity. This creates a significant detection gap, leading to long attacker dwell times and overwhelming security teams with low-quality alerts

Objectives :

- Design a deception-based defense framework to detect APTs.
- Generate reliable, high-fidelity alerts to reduce false positives.
- Profile attacker tactics, techniques, and procedures (TTPs) in a controlled environment.
- Map detected adversary behavior to the MITRE ATT&CK framework.

Methodology :

We propose a deception platform that with decoys and fake enterprise assets to lure attackers into a controlled environment. All interactions are monitored to generate high-fidelity alerts, which are then aggregated and processed in Splunk. Finally, these alerts are mapped to the MITRE ATT&CK framework for real-time analysis of adversary behavior.

Tools & Technologies :

- Deception Platform
- Splunk
- MITRE ATT&CK Framework
- Active Directory
- Simulated Enterprise Assets

Current Status :

- The initial deception environment has been established with the deployment of T-Pots.
- The Labshock platform has been installed to manage and orchestrate the deception assets.
- Splunk has been successfully integrated for centralized log collection and alert monitoring.
- The project is proceeding with the foundational setup and integration phase.

Expected Deliverables :

- A functional deception-based detection framework.
- A Splunk dashboard for real-time alert monitoring and threat analysis.
- Comprehensive documentation detailing the architecture and profiled APT TTPs.
- A final report demonstrating the framework's effectiveness in enhancing threat detection.

Innovation/Relevance :

Unlike traditional defenses by offering a proactive, behavior-based model. By luring attackers into a monitored deception environment, the solution forces them to expose their tactics. This provides security teams with actionable intelligence to quickly neutralize threats in critical infrastructure.

Keywords:

Advanced Persistent Threats (APT), Deception Technology, Critical Infrastructure, Threat Detection, MITRE ATT&CK, Splunk, T-Pot.